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1**Plasma and blood functions**

COMP

You should be able to: State the three functional categories of blood, transport, regulation, and protection, and name the major plasma proteins and the function of each.

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2**Hematocrit and formed elements**

COMP

You should be able to: Define hematocrit and interpret a value as normal or abnormal in the context of anemia and dehydration.

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3**Erythrocyte structure and hemoglobin**

COMP

You should be able to: Relate the biconcave shape and absent nucleus of the erythrocyte to its function and describe how the four heme groups of hemoglobin carry oxygen.

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4**Erythropoiesis and its regulation**

COMP

You should be able to: Trace the erythropoietin negative feedback loop from tissue hypoxia to increased red cell production and state the nutrients required.

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5**Red cell destruction and bilirubin**

COMP

You should be able to: Trace the fate of hemoglobin after red cell breakdown and explain how failure to clear bilirubin produces jaundice.

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6**Leukocytes**

COMP

You should be able to: Match neutrophils, lymphocytes, monocytes, eosinophils, and basophils to their defensive function and to the condition that raises each.

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Hemostasis

COMP

You should be able to: Order the three phases of hemostasis, vascular spasm, platelet plug formation, and coagulation, and identify the role of von Willebrand factor and thromboxane.

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Coagulation cascade

COMP

You should be able to: Compare the intrinsic and extrinsic pathways to the common pathway and identify the roles of thrombin, fibrin, calcium, and vitamin K.

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9**Clot limitation and fibrinolysis**

COMP

You should be able to: Explain how anticoagulants and plasmin limit and remove a clot and predict the effect of a drug that blocks vitamin K or platelet aggregation.

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10**ABO and Rh blood types**

COMP

You should be able to: Predict agglutination for any donor and recipient ABO and Rh combination and explain the mechanism of hemolytic disease of the newborn.

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11**Hematocrit and hemoglobin determination**

COMP

You should be able to: Determine hematocrit and hemoglobin from a sample or simulation and evaluate the result against reference ranges.

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12**Blood typing**

COMP

You should be able to: Perform simulated ABO and Rh typing and determine compatible donor types for the result obtained.

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13**Lymphatic return**

COMP

You should be able to: State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.

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14**Innate defenses**

COMP

You should be able to: Describe the physical, chemical, and cellular innate defenses and explain what makes them nonspecific and immediate.

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15**Inflammation and fever**

COMP

You should be able to: Order the events of the inflammatory response and explain how the cardinal signs arise and why fever aids defense.

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16**Adaptive immunity**

COMP

You should be able to: Compare humoral and cell mediated immunity by the cell responsible, the target, and the mechanism of elimination.

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17**Antibody structure and function**

COMP

You should be able to: Relate antibody structure to antigen binding and name the mechanisms by which antibodies neutralize or mark a target.

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18**Immune memory and immunization**

COMP

You should be able to: Compare the primary and secondary antibody responses and explain how active and passive immunization produce protection.

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Immune dysfunction

COMP

You should be able to: Classify allergy, autoimmunity, and immunodeficiency by what the immune system is doing wrong in each.

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